before crossing a threshold that makes total recovery impossible; the ease or difficulty in making a big change to the system; and how close to the threshold a system is currently.



A thick green scum of algae covers parts of Lonar Lake, in Maharashtra, India. Algae thrive in high-nutrient conditions, but decomposing algae consume oxygen, and depleted levels of oxygen lead to fewer fish surviving.

Changing states

According to Holling's view, resilience at the ecosystem level is enhanced by its populations not being too rigid—meaning that the components of the ecosystem can change. One example is the disappearance of most American chestnuts from forests in eastern North America, which was largely